Unit 04: Installing Software

CONTENTS

Objectives

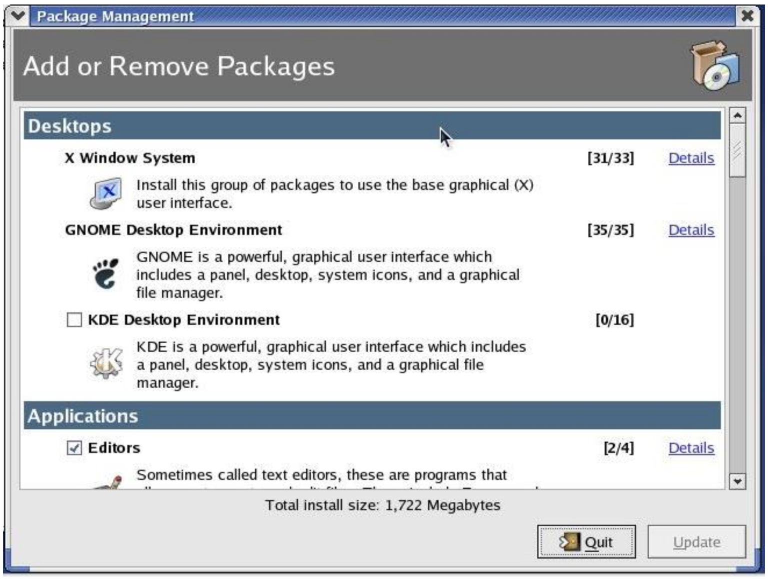
After studying this unit, you will be able to understand

- Understand RPM
- See the RPM Package Management tool
- Add and remove the packages
- Query the RPM package
- Install package in TAR format

Introduction

RPM stands for Redhat Package Manager. The RPM package manager is an open-source packaging system distributed under the GNU GPL. It runs on most Linux distributions and makes it easy for you to install, uninstall, and upgrade the software on your machine. RPM files can be easily recognized by their .rpm file extension and the 'package' icon that appears in your navigation window.The RPM package management is shown below.

4.1 RPM Package Manager



Benefits of using RPM: There are few reasons to use RPM.

- **Simplicity:** RPM is quite simple to use. The interface of RPM is very clear. The packages and the groups in RPM are very easy to locate. So, this is the remarkable feature of RPM.
- **Upgradability**: RPM interface is easy to upgrade. If a new package comes, it can be easily upgraded.
- **Manageability**: RPM interface is easily manageable. If we want to add or delete some packages using RPM, then it can be easily done. So, the manageability is one of the greatest feature of RPM interface.
- **Package Queries**: The packages are easily queried in RPM. By querying the packages, we can see which all packages are installed in the computer system.
- **Uninstalling**: It is very easy to uninstall a package or a group. If we don't need any package or its related extra group at some time, then it can be deleted at that time.
- **System Verification**: System verification can be easily done using RPM.
- **Security**: The RPM way of installing and installing packages is secure.

Ways to use RPM: RPM can be used in two different, yet complementary ways –

- From the desktop, using the GUI interface,
- From the command line.

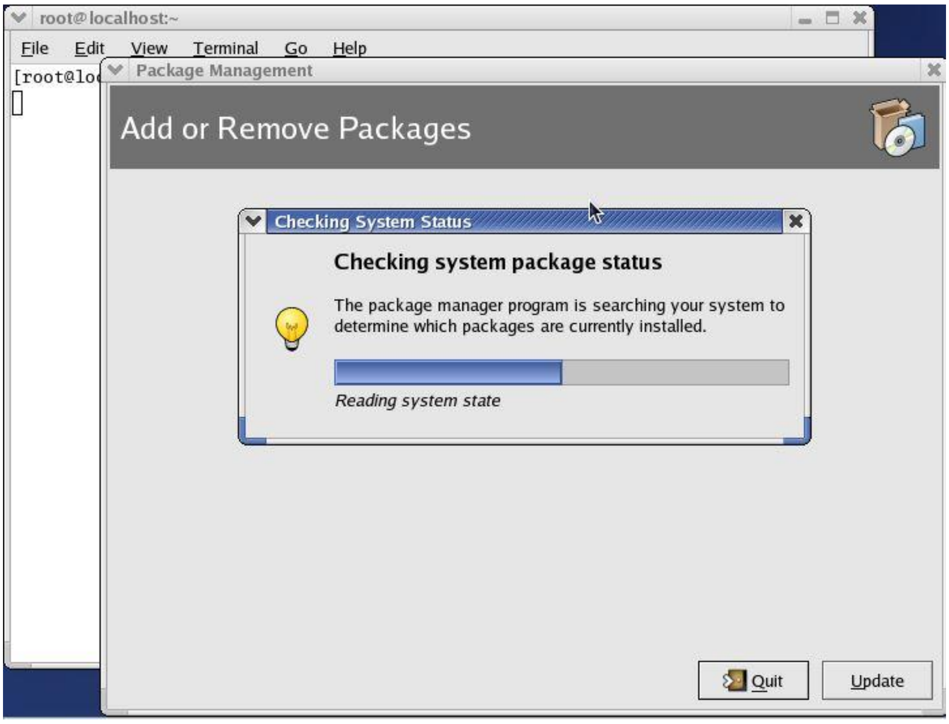
The RPM package management (GUI) tool

This tool is a graphical user interface (GUI) designed for the management of package installation and removal. The GUI allows us to add and remove packages at the click of a mouse.

Starting the RPM Package Management Tool: There are two ways to start RPM.

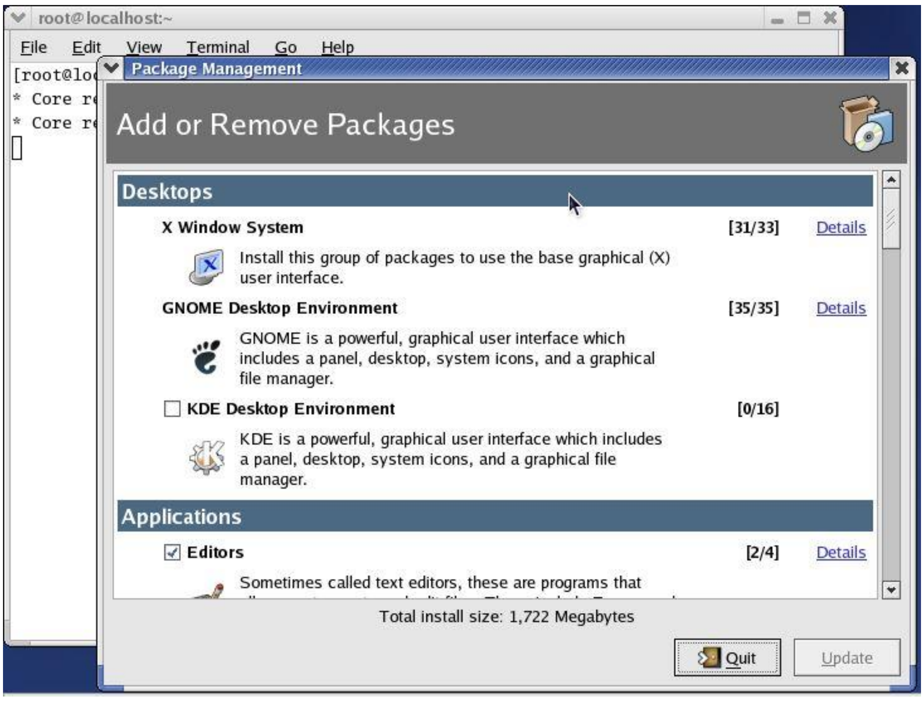
- Main Menu, select Main Menu | System Settings | Add/Remove Applications.
- $ redhat−config−packages

When you use either of the way, the next moment your screen will look like this. It checks the system package status.



The package management window shows all the packages and its group. The interface contains:

- Package category
- Package group
- Details link
- Number of packages installed/Out of total number of packages
- Summary of disk space required to install the package
- Update button
- Quit button



Package categories and groups: There are various package categories and groups available.

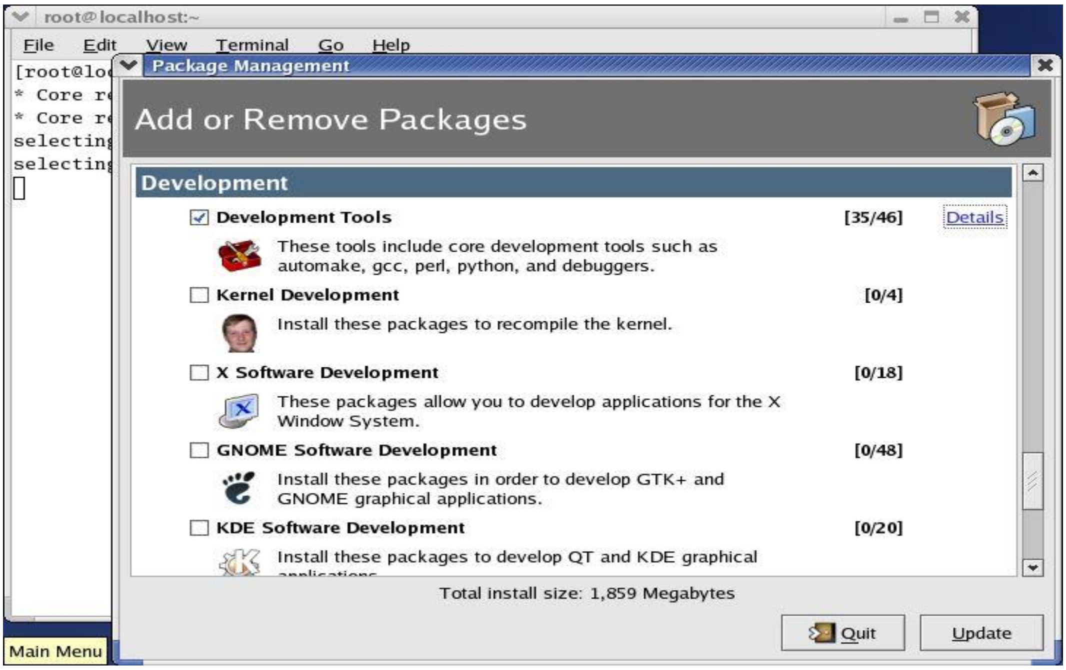
Standard and extra packages

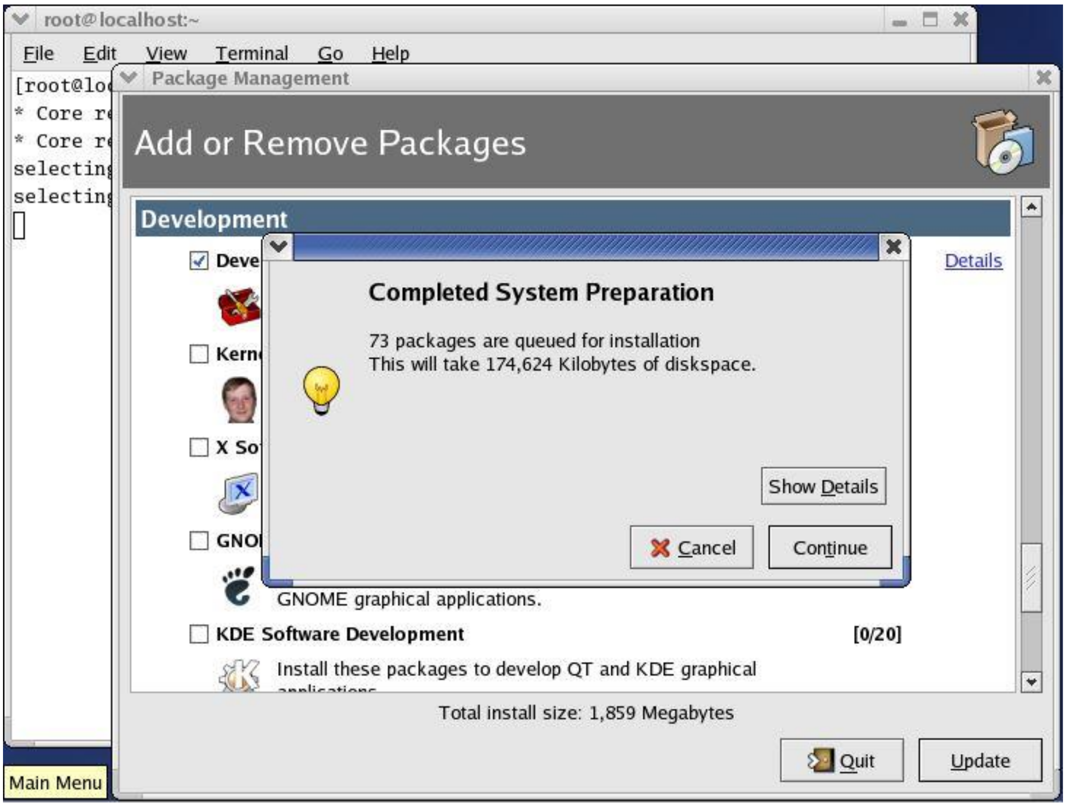
Each group may have standard packages and extra packages, or just extra packages. Standard packages are always available when a package group is installed − so you can't add or remove them explicitly unless the entire group is removed. However, the extra packages are optional so they can be individually selected for installation or removal at any time.

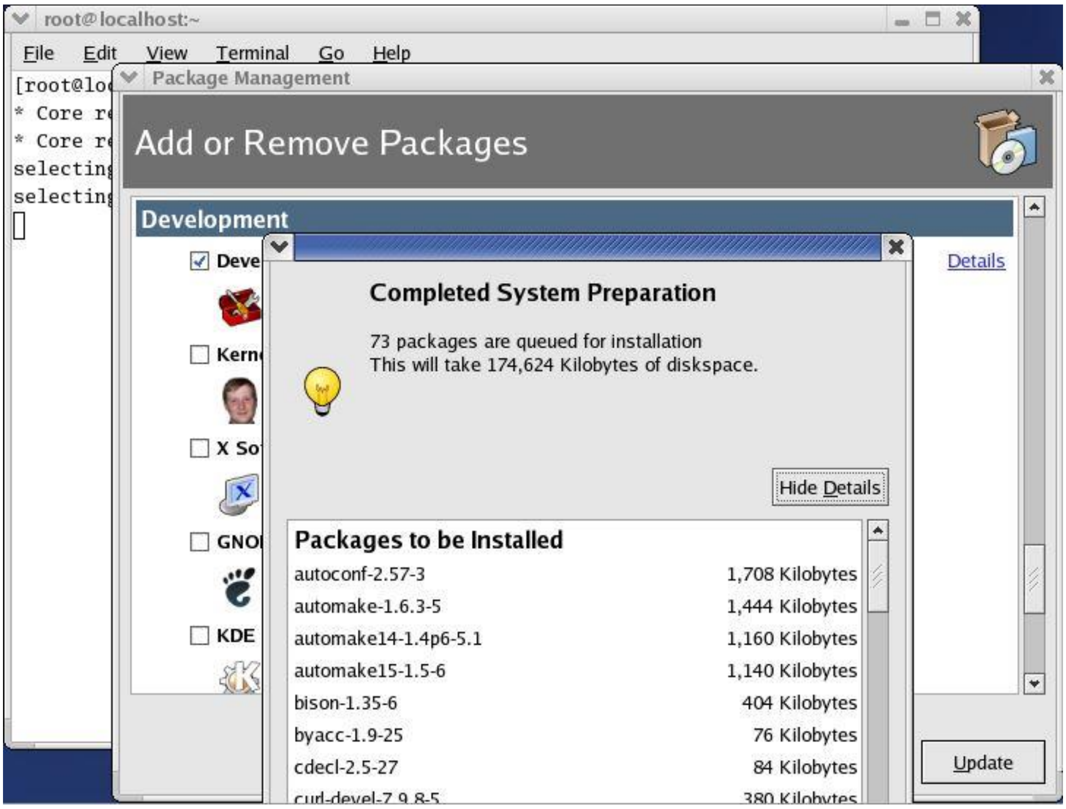
Package Category	Package Groups
Desktop	X Window System GNOME Desktop Environment KDE Desktop Environment
Applications	Editors Engineering and Scientific Graphical Internet Text−based Internet Office/Productivity Sound and Video Authoring and Publishing Graphics Games and Entertainment
Servers	Server Configuration Tools Web Server Mail Server Windows File Server DNS Name Server FTP Server SQL Database Server News Server Network Servers
Development	Development Tools Kernel Development X Software Development GNOME Software Development KDE Software Development
System	Administration Tools System Tools Printing Support

4.2 Adding and Removing Packages

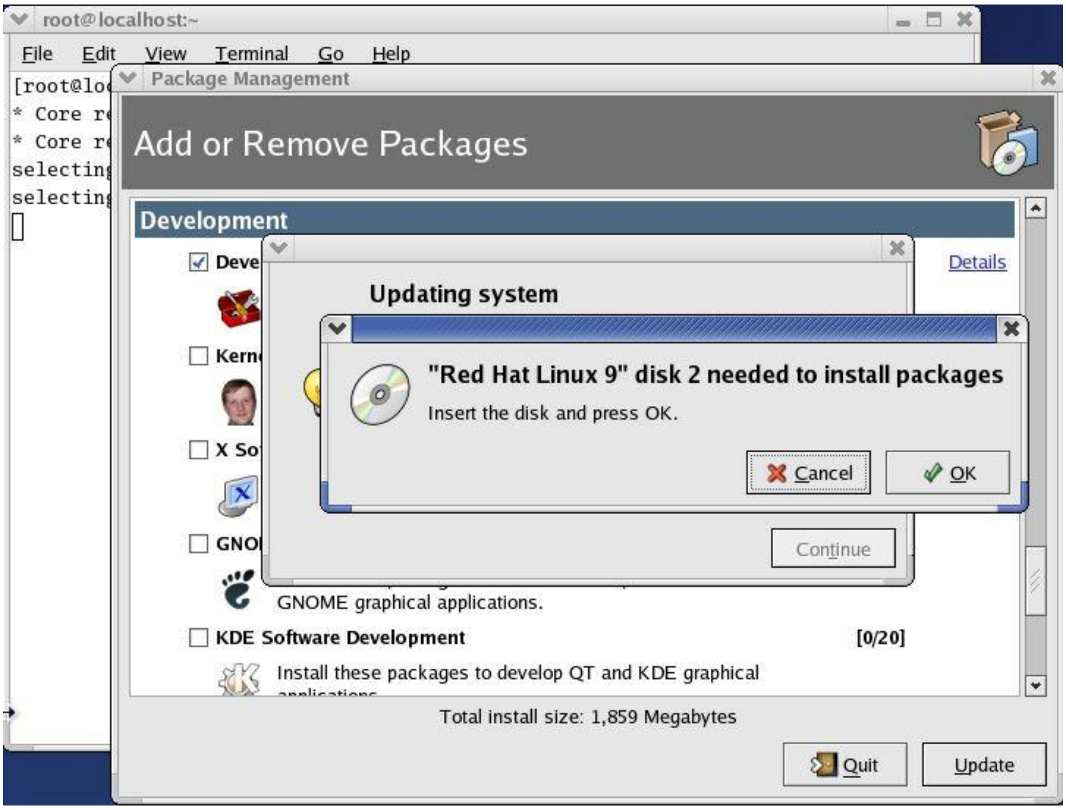
Installing new software from the package management tool is very simple. When we select any group using the RPM package management tool interface, it automatically selects the standard packages (if any) that are needed for the category as well as any dependent packages that it may have.We can customize the packages to be installed by clicking on the Details button. Once you've made your selections, click on the Update button on the main window. The package management tool will then calculate the disk space required for installing packages, as well as any dependencies, before displaying the following dialog:



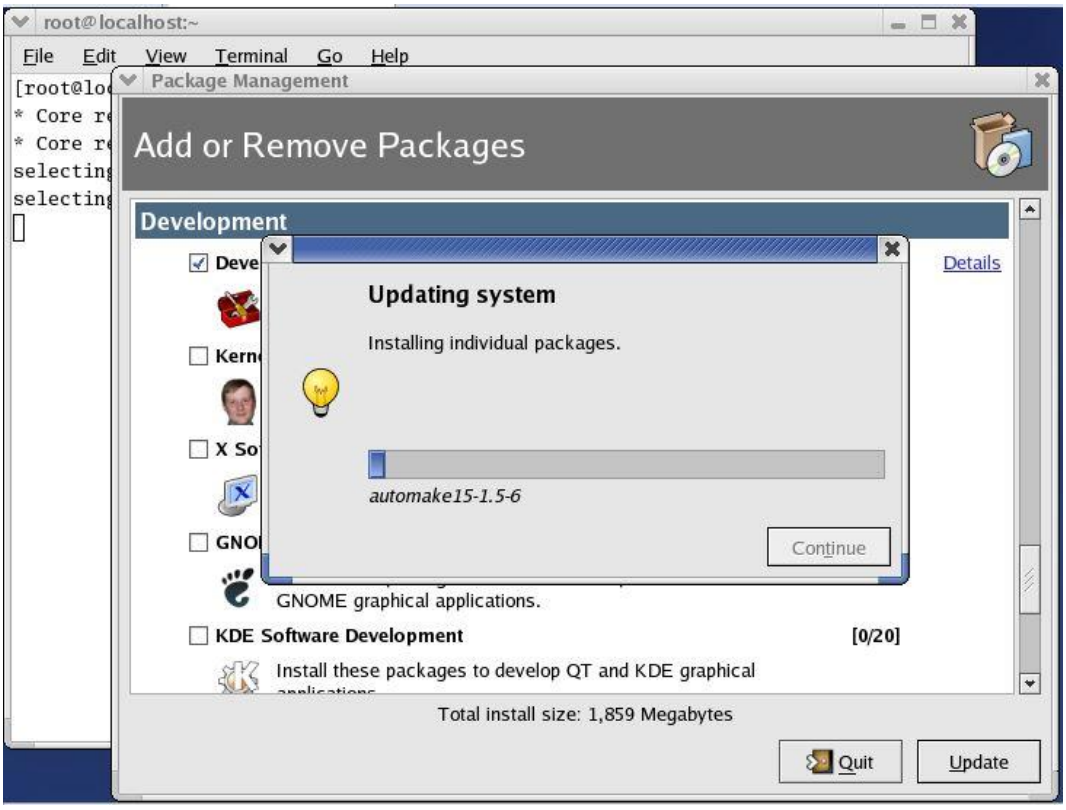




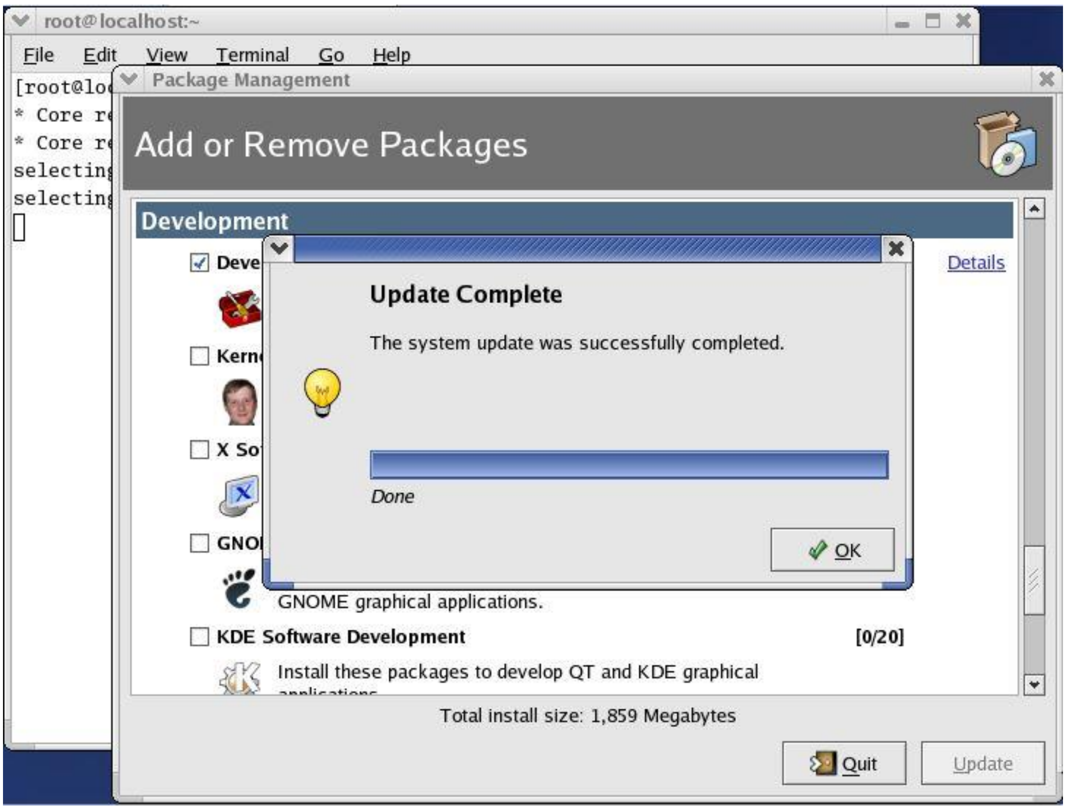
Use the required disk for updation of system.



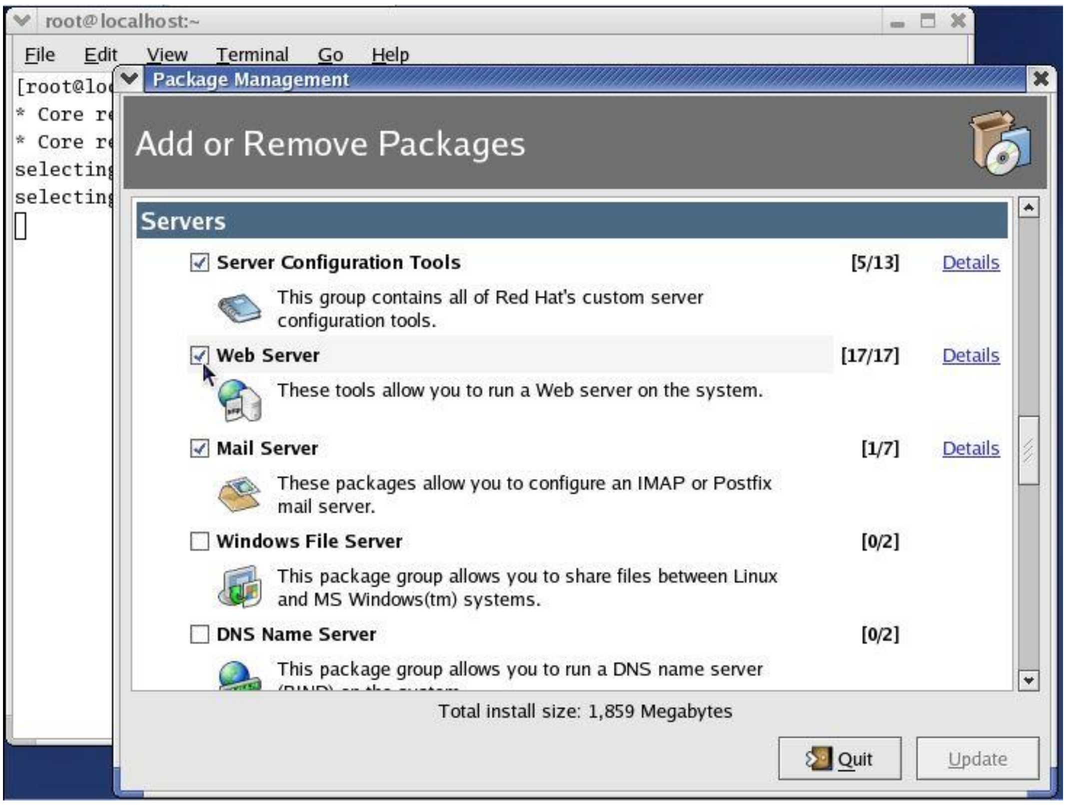
It will start updating the system.

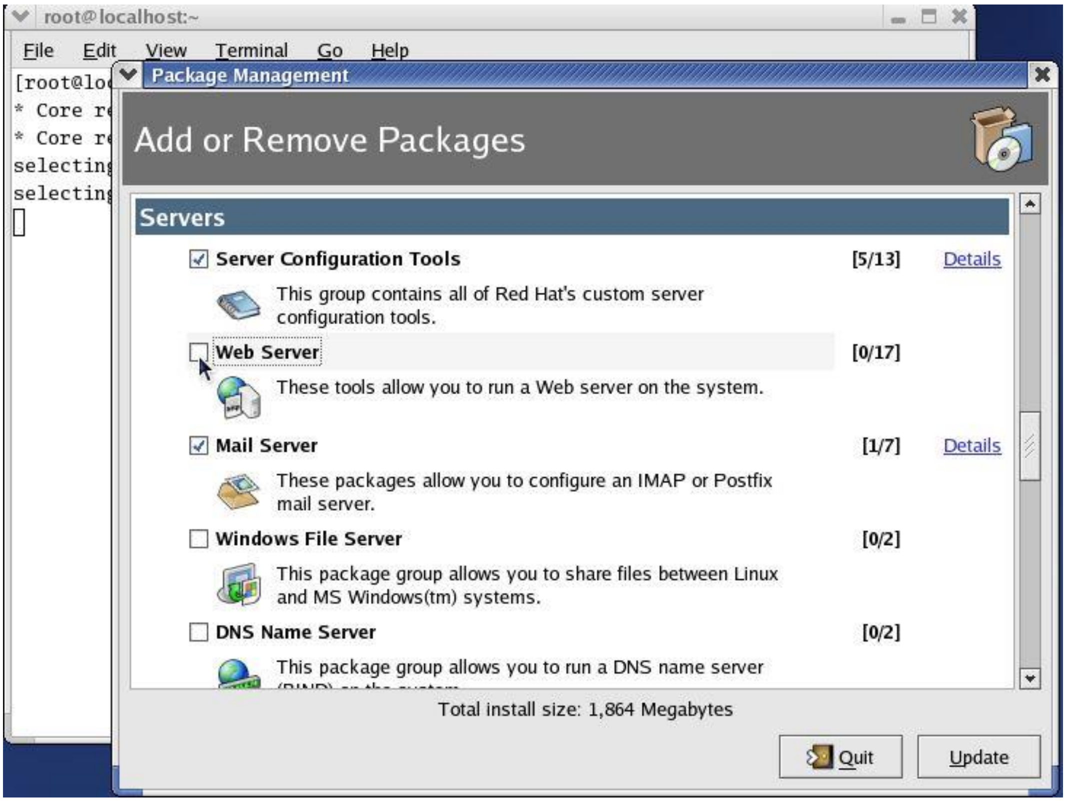


When the update is complete, it will show the updation complete message.

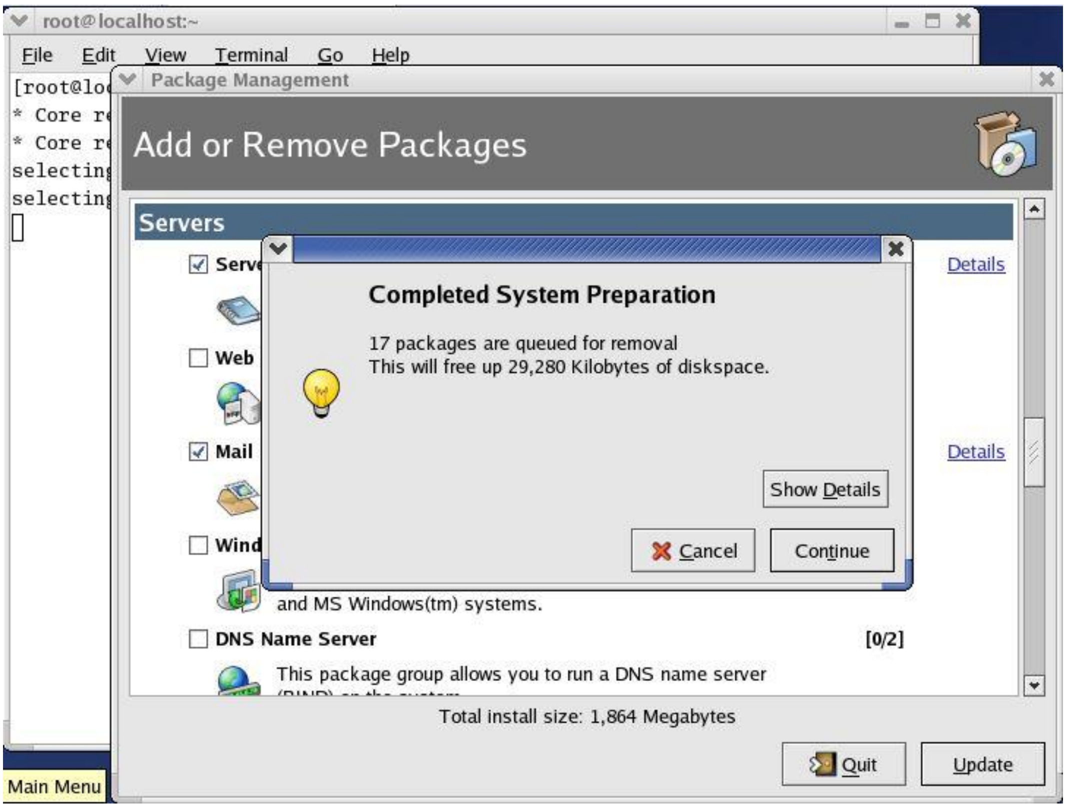


We can also delete the package if some package is no longer required. We just need to uncheck that package and click on update.

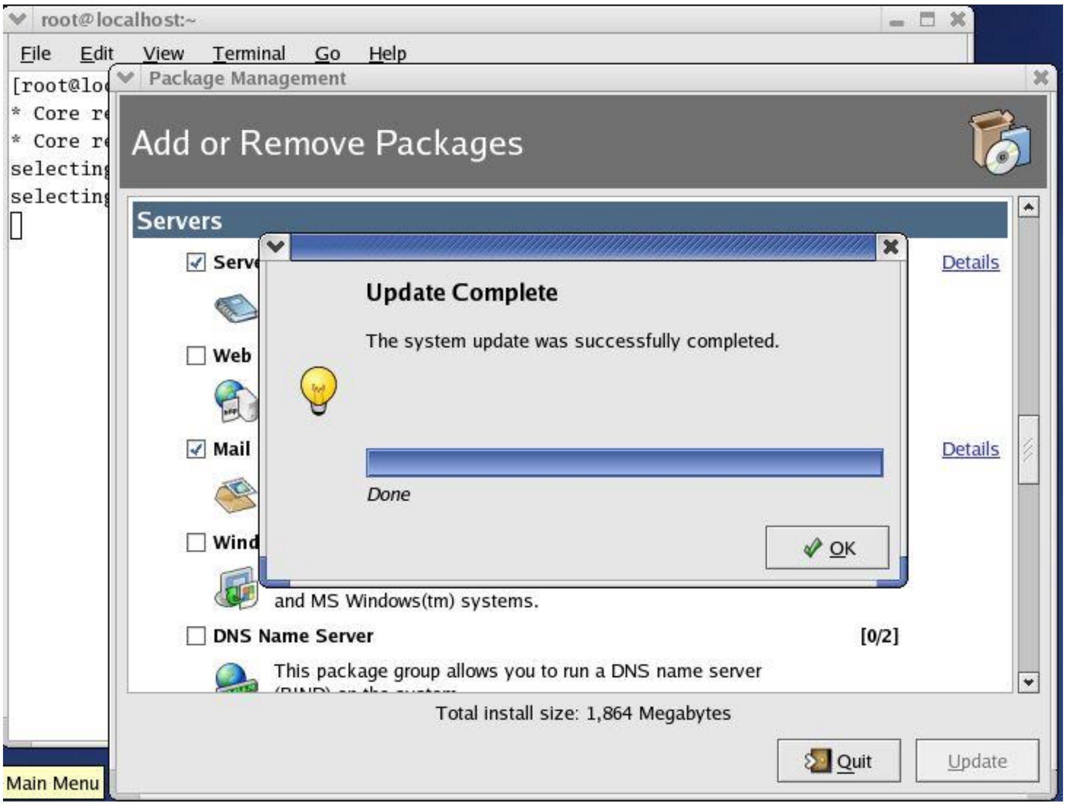




Then the packages will be placed in queue for deletion. Here the 17 packages are queued for removal. It will also show how much space will be freed after removal of packages. If the packages are correct for deletion (click on show details for confirmation), click on continue. It will remove the packages from the system.



When the update is complete. Click on ok.



4.3 RPM Command Line Tool

Up to now, we've talked about how to install and remove packages using Red Hat's graphical package management tool. While this tool is simple to use, it's lacking in functionality. For example:

- It cannot install packages using network, FTP, or HTTP connections.
- It does not show the location the files in a package are installed to.
- It lacks the capability to query for specific packages installed on the system.
- It does not provide full details of the RPM package − such as the vendor, build date, signature, description, and so on.
- It does not have a function to verify a package. That means it cannot compare information about files like size, MD5 sum, permissions, type, owner, and group installed from a package with the same information from the original package.
- It does not show all the packages available in a product. So you won't always know if you've got the whole thing.

4.4 Querying Packages

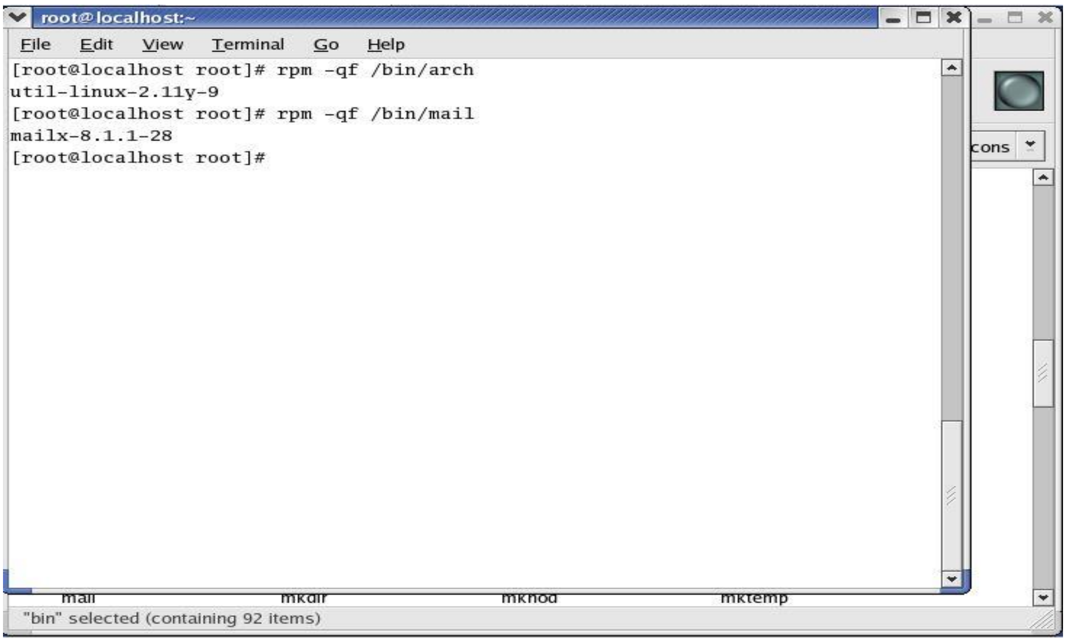
RPM keeps a record of all the packages installed on your system in a database. By querying the database, you obtain a complete list of all the packages that you've installed on your system. Should you want to, you can then go further and query each individual package for more details about itself.

The syntax for a basic query is as follows:

- rpm −q [options] <filename>

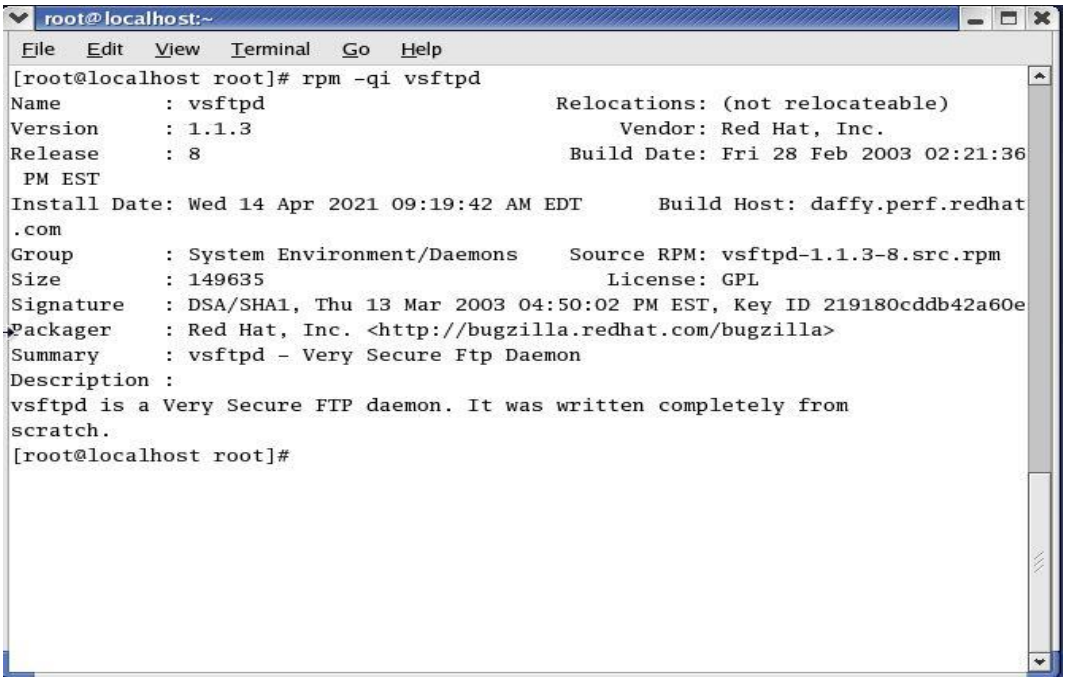
You have list of files, and you would like to find out which package belongs to these files.The syntax is:

- rpm –qf<filename>



You have installed an rpm package and want to know the information about the package. The syntax is:

- rpm –qi <package _name>

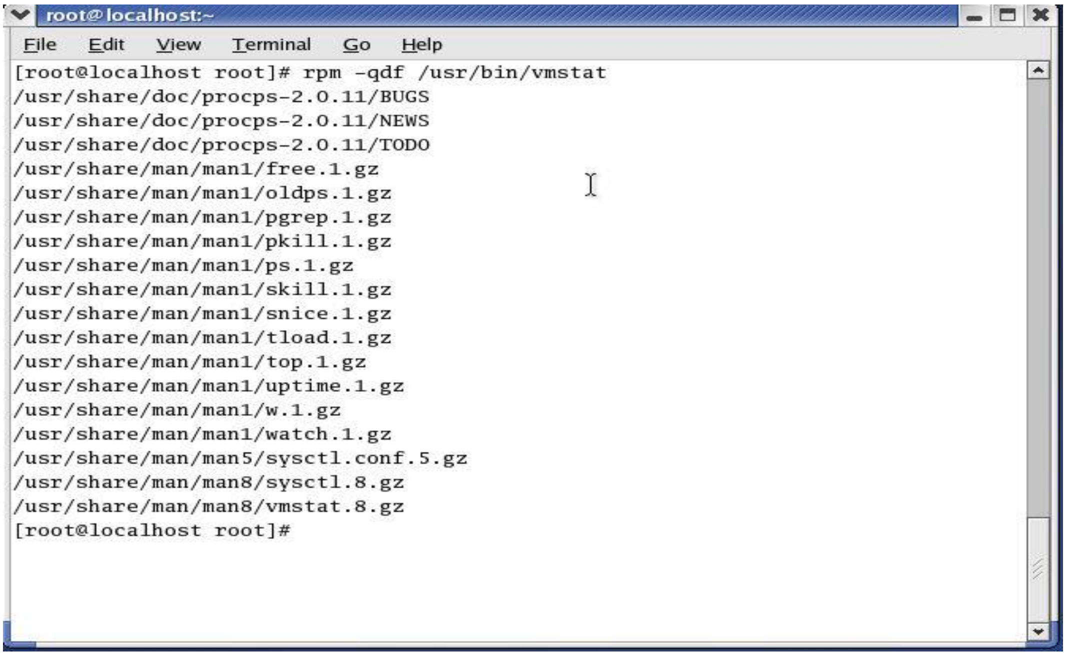


You have downloaded a package from the internet and want to know the information of a package before installing.The syntax is:

- rpm –qip<package_name>

To get the list of available documentation of an installed package.The syntax is:

- rpm –qdf<package_name>



4.5 Package Installation in TAR Format

TAR stands for tape archive.An archive is nothing, but you bundle (put) many files together into a single file on a single tape or disk.The tar program combines multiple files into a single large file. It is separate from the compression tool, so it allows you to select which compression tool to use or whether you even want compression.A tar ball is a (usually compressed) archive of files, similar to a Zip file on Windows or a Sit on the Mac. Tar balls come in files that end in .tar, .tar.gz, .tgz, or something along these lines.

- Structure of the tar command

  [root@localhost /root]# tar [commands and options] filename

- Option for tar
  -c Create a new archive.
  -t View the contents of an archive.
  -x Extract the contents of an archive.
  -f Specify the name of the file (or device) in which the archive is located.
  -v Be verbose during operations.
  -z Use gzip to compress or decompress the file.
  -u To upgrade the tar file

To create a tar file

[root@localhostmohit]# tar -cvf filename.tar directory [path of files]

In this example, filename.tar represents the file you are creating and directory/file represents the directory and file you want to put in the archived file.

Where,

-c : Create a tar ball.

-v : Verbose output (show progress).

-f : Output tar ball archive file name.

-x : Extract all files from archive.tar.

-t : Display the contents (file list) of an archive.

To View a Tar Ball

- It Contains (list file inside a tar ball)
- Type the following command:

  tar -tvf/tmp/data.tar **ar Ball**

To Extract a Tar Ball

Type the following command to extract /tmp/data.tar in a current directory, enter:

tar -xvf/tmp/data.tar

Summary

- RPM files can be easily recognized by their .rpm file extension and the 'package' icon that appears in your navigation window.
- There are various benefits of using RPM for package installation and deletion: simplicity, upgradability, manageability, easy uninstallation, system verification and high security.
- The RPM package management tool contains: the button for package category, package group, details link, number of packages installed/Out of total number of packages, summary of disk space required to install the package, update button, and quit button.
- Each group may have standard packages and extra packages, or just extra packages.
- We can customize the packages to be installed by clicking on the Details button. Once you've made your selections, click on the Update button on the main window.
- RPM keeps a record of all the packages installed on your system in a database.

- The tar program combines multiple files into a single large file. It is separate from the compression tool, so it allows you to select which compression tool to use or whether you even want compression.

Keywords

- **RPM:**The RPM package manager is an open-source packaging system distributed under the GNU GPL.
- **Standard packages**: These are always available when a package group is installed − so you can't add or remove them explicitly unless the entire group is removed.
- **Extra packages**:These are optional so they can be individually selected for installation or removal at any time.
- **Archive:**An archive is nothing, but you bundle (put) many files together into a single file on a single tape or disk.
- **Tar ball**: A tar ball is a (usually compressed) archive of files, like a Zip file on Windows or a Sit on the Mac. Tar balls come in files that end in .tar, .tar.gz, .tgz, or something along these lines.

Self Assessment

1. TAR stands for

A. Tour Archive
B. Tape Archive
C. Tape Assistance
D. Tour Assistance

2. Under development tools, what can be installed using RPM package management tool?

A. KDE Software development
B. GNOME Software development
C. X Software development
D. All of the above

3. While graphical interface of RPM package management tool can install/remove/update the packages, but it still lacks which functionality.

A. It cannot install packages using network, FTP, or HTTP connections.
B. It does not show the location the files in a package are installed to.
C. Both above
D. None of the above

4. The RPM package management tool is a ______

A. Graphical interface
B. Textual interface
C. Not an interface
D. None of the above

5. Using RPM package management tool, we can install
A. Web server
B. Mail server
C. DNS name server
D. All of the above

6. We can which packages are installed/ not installed by clicking on
A. Update button
B. Quit button
C. Details link
D. None of the above

7. Which of these packages are always available when a package group is installed?
A. Standard packages
B. Extra packages
C. Grouped packages
D. None of the above

8. Which of the buttons are available on the tool interface?
A. Update button
B. Quit button
C. Both update and quit buttons
D. None of the above

9. The slash (/) in the interface of RPM package management tool represents:
A. Total number of packages/ Number of packages installed
B. Number of packages installed/ total number of packages
C. Package category/ package group
D. Package group/ package category

10. What is the extension of the RPM file
A. .txt
B. .doc
C. .rpm
D. .pdf

11. How can we start RPM package management tool?
A. $redhat-config-services
B. $redhat-config-packages
C. $redhat-config-management
D. None of the above

12. What are the benefits of using RPM?

A. Package queries
B. System verification
C. Security
D. All of the above

13. With RPM, it is easy to ______ softwares on the computer system.
A. Install
B. Uninstall
C. Upgrade
D. All: install, uninstall and upgrade

14. From ___________________, it is possible to install packages network, FTP or HTTP connections.
A. Command line
B. Graphical interface
C. Both command line and graphical interface
D. None of the above

15. Which of these commands is used for querying the information about a package after installation?
A. rpm –qi<filename>
B. rpm –qu<filename>
C. rpm –qr<filename>
D. None of the above

Answers for Self Assessment

1.	B	2.	D	3.	C	4.	A	5.	D
6.	C	7.	A	8.	C	9.	B	10.	C
11.	B	12.	D	13.	D	14.	A	15.	A

Review Questions:

1. What is RPM? Write the ways and benefits of using RPM.
2. What is RPM package management tool? How can we start it? Explain some details about its interface.
3. How can we add and remove the packages? Explain.
4. What is RPM command line tool? Write its benefits.
5. How can we query a package? Write syntax.
6. Explain the package installation in TAR format. How can we create, view and extract a tar ball?

Further Readings

Mark G. Sobell, A Practical Guide to Fedora and Red Hat Enterprise Linux, Fifth Edition, Pearson Education, Inc.

Web Links

https://www.redhat.com/sysadmin/create-rpm-package